

Toward Efficient Wireless Federated Learning via Decoupling Over-the-Air Model Aggregation and Client Selection

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Abstract—Federated learning (FL) is revolutionizing machine learning by enabling multiple decentralized clients to collaboratively train a shared model. In mobile scenarios, client devices exchange model parameters with a central server via wireless channels. However, designing efficient wireless FL (WFL) is challenging due to limited energy and channel capacity. To fully utilize communication resources, over-the-air (OTA) computation has been introduced, allowing direct aggregation of analog parameter signals. However, it conceals clients' information from the server, making advanced client selection strategies, e.g., cluster-based client selection, and sparsification compression algorithms like TopK inapplicable. To address these limitations, we propose the WFL with Voting-based Clustering (WFL-VC) algorithm, which can integrate advanced client selection and the TopK model sparsification algorithm with OTA computation. WFL-VC consists of two phases: 1) Phase 1: clients vote on significant parameters based on local models, allowing the server to select clients and identify the global Topk parameters; and 2) Phase 2: the selected clients upload model update parameters with globally aligned indices for over-the-air computation at the server. By combining OTA computation with cluster-based client selection and TopK sparsification, WFL-VC substantially reduces the energy consumption of WFL. Extensive experiments on real-world datasets show that WFL-VC outperforms competitive baselines while consuming considerably less energy.

Index Terms—Client selection, data heterogeneity, over-the-air computation, voting-based clustering, wireless federated learning (WFL).

I. INTRODUCTION

WITH the rapid development of the Internet of Things (IoT) and Artificial Intelligence (AI), cellular networks are transitioning from the concept of connecting everything in

5G to connecting intelligence in 6G [1], [2]. This paves the way for harnessing the massive data generated from the ever-increasing edge devices (such as vehicles, IoT devices, and smartphones) to train advanced models. However, adhering to the traditional cloud-centric training mode, which requires sharing local data with a central unit, may raise privacy concerns. Recently, the federated learning (FL) paradigm has emerged as a privacy-preserving approach, enabling collaborative training among decentralized edge devices without sharing private data [3]. A parameter server (PS) is deployed to manage training by collecting model updates from multiple edge devices and aggregating them into a global model.

FL is profoundly transforming machine learning and has been widely adopted in various contexts, including mobile computing systems, recommender systems [4], and IoT systems [5]. Among these application scenarios, conducting FL in wireless environments is particularly challenging due to the limited energy and communication resources of wireless devices [6], [7]. Distributed computing in FL can be realized in digital fashion via traditional multiple-access channel (MAC) technologies [8], such as TDMA, OFDMA, and CDMA. Under MAC transmission, clients first digitize their model updates and then transmit them over orthogonally allocated channel resources to the PS. The server can carry out the aggregation process only after decoding all participating clients' model updates. However, the limited channel resources available to each device restrict the quality of model updates that clients can transmit. Moreover, the communication-and-computation approach used when adopting MAC for model updates transmission in FL further results in significant bandwidth and latency overhead [9].

Complex neural networks are increasingly used in advanced machine learning applications deployed on wireless devices, imposing a significant burden as the number of parameters transmitted by each device scales proportionally with the neural network size [10]. As reported in [11], when training a deep network for image classification tasks (e.g., ResNet-50 model), approximately 100 MB parameters have to be transmitted over 100 Mbps wireless uplinks, typically consuming 30 J of battery energy from a mobile device (e.g., Huawei Mate 10 with a 4000 mAh, 53 kJ battery) [12]. This is similar to the energy required for a single local training iteration on a high-performance data center GPU (e.g., NVIDIA Tesla V100, which consumes approximately 30 J per

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iteration on the ImageNet dataset). Thus, transmitting large volumes of model updates can overwhelm communication capacity, while continuous local training rapidly consumes power, quickly depleting the limited battery capacity.

To fully utilize the communication capacity of each client device, over-the-air computation, also called over-the-air (OTA) aggregation, has been introduced in AirFL [13]. AirFL exploits the superposition property of transmitted analog waveforms [14] to aggregate model updates from multiple clients simultaneously. Specifically, multiple clients transmit their model updates over the same time-frequency resources, resulting in direct waveform superposition before the signals reach the PS. While this allows each client to fully utilize the available communication capacity and hence accelerates WFL, it also means that with OTA aggregation, the PS receives only aggregated model information and cannot distinguish or analyze individual client characteristics. Consequently, client selection strategies [15], [16], [17], which rely on analyzing individual client information, become infeasible in AirFL. Some existing studies employ only simple random client selection in AirFL [9], [18] since these frameworks rigidly couple OTA model aggregation with client selection. This restriction also explains why energy efficiency improvements for WFL through advanced client selection cannot be directly extended to AirFL.

Furthermore, training large models in AirFL can incur communication overhead unless the transmitted model updates are compressed. To reduce this overhead, model compression techniques, such as quantization and sparsification, are typically used. However, integrating model compression techniques into AirFL is nontrivial. Model updates are transmitted as an analog waveform, which cannot be directly quantized by quantization schemes such as QSGD [19], ClusterGrad [20]. Moreover, when applying sparsification compression such as TopK [21], only the most significant parameters from each client's model updates are selected for transmission. This creates a conflict with AirFL's requirement for unified index alignment across all transmitted parameters to perform over-the-air aggregation.

To improve WFL training efficiency by addressing the above limitations, we propose WFL with Voting-based Clustering (WFL-VC), an algorithm that integrates advanced client selection strategies and TopK model compression with OTA aggregation. WFL-VC operates in two phases per training round. In Phase 1, each client performs local training and then transmits votes for significant model updates based on its local model. To minimize voting overhead, each parameter is represented by a single bit, compactly transmitted over digital communication channels. The PS collects votes from clients to first estimate client distances, which serve as criteria for clustering clients into multiple groups, and second, to aggregate the votes to identify indices of global significant model updates. To reduce energy consumption, WFL-VC selects the most representative clients from each cluster to participate in OTA aggregation. Finally, the PS distributes both the indices of global significant model updates and the selected clients list to all clients. In Phase 2, the selected clients transmit their model updates via analog waveforms. The main contributions of this work are summarized as follows.

- 1) We investigate the limitations of wireless FL (WFL), highlighting the tradeoff between enabling client clustering, TopK model compression, and the inherent constraints posed by over-the-air computation.
- 2) We propose WFL-VC, a novel algorithm that seamlessly combines client selection strategies and TopK model compression with OTA aggregation in WFL. This integration simultaneously optimizes communication and energy efficiency in WFL.
- 3) We evaluate the performance of WFL-VC on the MNIST and CIFAR-10 datasets under non-IID data distributions. We further analyze the performance across different hyperparameter combinations to verify the robustness of the proposed algorithm.

The remainder of this paper is organized as follows. Section II reviews the most relevant research and identifies the gaps addressed by this study. In Section III, we delve into the fundamental principles of FL and model clustering. Section IV presents the proposed algorithm, followed by experimental evaluation on real datasets in Section V. Finally, Section VI concludes our work and outlines directions for future research.

II. RELATED WORK

In recent years, extensive research has explored FL applications in various domains, including recommendation systems, healthcare, and industrial IoT [22]. Under the FL paradigm, the PS coordinates model training by exchanging model updates with clients over digital or analog communication. However, a key challenge in deploying FL in systems composed of wireless devices lies in improving efficiency and reducing participation rounds, as these devices typically have limited communication capacity and constrained battery life [12]. In this section, we discuss related works from two perspectives: WFL and client selection in FL.

A. Wireless FL

1) *FL With Digital Communication:* When considering digital communication for WFL, orthogonal multiple-access schemes distribute bandwidth resources, allowing multiple clients to transmit model updates to PS at the same time. When numerous clients participate in WFL training, limited bandwidth allocation per client device leads to significant communication delays and restricts system scalability [23], [24]. Transmission via orthogonal multiple access prevents the PS from simultaneously receiving and aggregating clients' model updates [25], imposing transmission delays and limiting the FL training efficiency.

2) *FL With Analog Communication:* Over-the-air aggregation has been proposed [13] to enhance the efficiency of model updates transmission from clients to the PS in WFL. With AirFL adopting OTA aggregation, clients simultaneously transmit their model updates as analog waveforms that are directly aggregated over the air before reaching the server. This approach permits each client to fully exploit its available communication capacity. Nonetheless, the transmitted model updates in AirFL are affected by wireless channel conditions during transmission. In [26], the authors analyze the effects of aggregation error on AirFL and investigate

how to optimize transmission power control to mitigate data aggregation errors. The Air-FedAvg algorithm proposed in [27] implements federated averaging, where OTA-based model aggregation occurs after multiple local training iterations. It aims to improve training accuracy by jointly optimizing the denoising factor and the transmission power of client models. In [28], AirFL is studied over time-varying wireless channels, where multiple antennas are used to combat the effects of channel fading. Nevertheless, training large models in AirFL remains inefficient due to the difficulty of achieving high compression rates. Since communication cost in FL is largely determined by the transmitted model size, model compression has become an effective approach to mitigate communication overhead.

3) *Model Compression*: Compression algorithms reduce the amount of traffic transmitted per global iteration, thereby improving communication efficiency in FL. In digital communication, model updates are encoded before transmission, with quantization techniques used to map model updates to a small set of discrete values. In AdaGQ [29], model updates are compressed through a quantization algorithm that adaptively adjusts the quantization level in response to variations in gradient norms and clients' communication capability. MUCSC [30] introduces a model update compression algorithm based on soft clustering that computes centroids according to the model updates distribution. Under a fixed code length, each model update is represented by a cluster centroid and cluster ID. In [31], an adaptation framework is proposed that adjusts the compression rate to maximize model accuracy. Apart from quantization-based methods, sparsification techniques offer an alternative direction for reducing communication overhead by limiting the number of model updates transmitted. TopK [21] is a widely adopted sparsification algorithm used in FL that selects only k model updates with the largest magnitudes for transmission. Lu et al. [32] propose an adaptive gradient compression algorithm that sets different compression ratios based on specific client data metrics.

However, compression algorithms designed for digital communication are incompatible with analog communication systems. In analog communication, model updates are transmitted as waveforms that correspond to the model update magnitude. Hence, quantization algorithms are unsuitable for AirFL because mapping model updates to discrete values disrupts the direct over-the-air aggregation achieved via waveform superposition. When using sparsification algorithms, clients select local significant model updates, which correspond to different indices across clients. As a result, over-the-air aggregation becomes infeasible because it requires the indices of model updates transmitted by different clients to be aligned.

Given the strict requirement for index alignment to enable over-the-air aggregation, developing effective model compression techniques for analog communication remains a challenging problem with limited existing works. In [33], a one-bit compressive sensing analog aggregation algorithm (OBCSAA) is developed for AirFL, incorporating TopK sparsification. To solve the sparsified model update alignment problem before OTA aggregation, OBCSAA applies

random linear projection (as in [34]) for dimensionality reduction, followed by 1-bit quantization for further compression. Despite achieving compression, the reduced dimensionality remains linearly proportional to the FL system scale, making OBCSAA inefficient for training large models in large-scale FL systems. To address this limitation, Su et al. [18] propose WFL-VTopK, a voting-based algorithm for AirFL. It overcomes the challenge of implementing TopK sparsification in OTA aggregation by enabling clients to collaboratively determine a consensus set of global indices. This ensures that all clients follow a unified indexing scheme, allowing direct aggregation of sparsified model updates. Additionally, it jointly optimizes the power allocation for both transmission phases (voting and training) and denoising to enhance efficiency. However, these existing approaches typically assume either random client selection or full client participation, neglecting the potential efficiency gains from optimized client selection in WFL. To bridge this gap, we introduce a novel framework that integrates advanced client selection strategies and model compression with OTA aggregation, optimizing WFL from both the communication and energy perspectives.

B. Client Selection in FL

In FL systems, both resource capacity and data distribution are heterogeneous across decentralized clients. This inherent heterogeneity makes the design of advanced client selection strategies critical for optimizing system efficiency [35]. A simple full client participation strategy will unnecessarily exhaust clients' resources, while a random selection strategy fails to distinguish clients' different contributions to FL. Both approaches result in low system efficiency. Cluster-based client selection is a widely adopted approach in FL, where clients are grouped according to either system capacity or data distribution similarity before being selected for participation. For instance, after clustering clients, Fraboni et al. [36] assign each client a participation probability based on the characteristics of its cluster. Xu and Wan [37] cluster clients based on computation and communication delay, followed by hierarchical and asynchronous model aggregation. Wolfrath et al. [38] optimize client selection by considering both model accuracy and training latency, while Sun et al. [39] prioritize clients based on computational capacity. Beyond client selection, client clustering is also used to enhance personalized model training. Sattler et al. [40] group clients with similar data distributions to train personalized models, addressing the challenge of data heterogeneity. In clustered FL (CFL), clients are clustered based on the cosine similarity of their gradient updates, utilizing K -means clustering [41]. Similarly, Liang et al. [42] propose a more lightweight approach by computing client similarity using only a single layer of the trained model instead of the entire model. Nevertheless, existing client clustering algorithms designed for FL are not applicable to AirFL, as the server cannot access individual clients' information during over-the-air aggregation to estimate client similarity. Our work addresses this challenge by designing a novel voting-based clustering method.

TABLE I
NOTATION LIST

Notation	Meaning
$\mathbf{w}_t$	Global model parameters in the t th global iteration.
$p_{i,1}, p_{i,2}$	Energy factor of client i in upload phase 1/2.
ϵ	Denoising factor.
$\mathbf{U}_t^i$	Model updates of client i in the t th global iteration.
$\mathbf{e}_t^i$	Residual model updates do not align with global index vector.
k	Number of votes.
$\mathbf{v}_t^i$	Index array for client i in the t th global iteration.
$\mathbf{v}_t$	Global index array in the t th global iteration.
$P_{max,t}^i$	Maximum power budget for client i in t th global iteration.
h_i	Channel coefficient between client i and the PS.
Ω_i	Uploading rate of client i .
$\mathcal{S}_t$	Selected clients in the t th global iteration.
M	Number of clusters.
C^m	Cluster id of m th cluster.
$\mathbf{o}^m$	Centroid of m th cluster.
$\mathcal{G}^m$	Group of clients at the m th cluster.
$\rho^{\mathcal{G}^m}$	The selection probability of all clients in m th group.
λ	Decay factor of the selection probability.
a	Threshold integer to eliminate less voted model parameters.
b, c	Weight parameters of the participant selection method $\kappa^m(i)$.
$\mathcal{S}_t^m$	Selected clients from m th group to represent m th cluster.
ψ_t^i	Selected client indication.
n	Number of selected clients to represent each cluster.

III. PRELIMINARIES

A. Federated Learning

In FL, a single PS is deployed to coordinate N decentralized client devices, where each device i has a local dataset $\mathcal{D}_i$. FL aims to train a model $\mathbf{w} \in \mathbb{R}^d$ with dimension d by minimizing the global loss function $F()$

$$\begin{aligned} \min_{\mathbf{w} \in \mathbb{R}^d} F(\mathbf{w}) &= \frac{1}{N} \sum_{i=1}^N F_i(\mathbf{w}, \mathcal{D}_i) \\ F_i(\mathbf{w}, \mathcal{D}_i) &= \frac{1}{|\mathcal{D}_i|} \sum_{\xi \in \mathcal{D}_i} f(\mathbf{w}, \xi) \end{aligned} \quad (1)$$

where $f(\mathbf{w}, \xi)$ is the loss function for evaluating the model $\mathbf{w}$ with the given sample ξ and $|\mathcal{D}_i|$ denotes the size of client i dataset. In FedAvg [3], the PS distributes the global model to the participating clients at every iteration t . Initially, clients download $\mathbf{w}_t$ from PS to establish their local model as $\mathbf{w}_{t,0}^i = \mathbf{w}_t$, then perform E rounds of local iterations using the batch gradient descent algorithm as

$$\mathbf{w}_{t,j+1}^i = \mathbf{w}_{t,j}^i - \eta_t \nabla F_i(\mathbf{w}_{t,j}^i, \mathcal{B}_{t,j}^i) \quad (2)$$

where $j = 0, 1, \dots, E-1$ and $\mathcal{B}_{t,j}^i$ is a batch data of size $|\mathcal{B}_{t,j}^i|$ sampled from $\mathcal{D}_i$. Once the local training is completed, each client will upload model updates $\mathbf{U}_t^i = \mathbf{w}_{t,0}^i - \mathbf{w}_{t,E}^i$ (rather than raw data) to the PS for aggregation. At the PS, the received model updates are aggregated as

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{1}{N} \sum_{i=1}^N \mathbf{U}_t^i. \quad (3)$$

After refining the global model via (3), the PS proceeds to the next global iteration, and the process repeats until it reaches the specified number of outer iterations T .

B. Model Clustering Based on Distance

In practical scenarios, data generated by each client follows a different distribution. Local models trained on similarly distributed data are expected to exhibit analogy, and the data within each cluster should share the same distribution. On the one hand, to perform clustering, a distance metric is required to measure the divergence between different clients' models. One such metric is cosine similarity, which defines the distance between two vectors. In general, let $\mathbf{v}^i$ and $\mathbf{v}^j$ represent two vectors owned by clients i and j , respectively. The distance between i and j based on $\mathbf{v}$ can be defined as follows:

$$\text{dis}(\mathbf{v}^i, \mathbf{v}^j) = 1 - \frac{\mathbf{v}^i \cdot \mathbf{v}^j}{\|\mathbf{v}^i\| \cdot \|\mathbf{v}^j\|}, \quad i \neq j. \quad (4)$$

On the other hand, to evaluate the quality of the clustering result, a commonly used metric is the average silhouette coefficient π [43], which can be expressed as follows:

$$\pi = \frac{1}{N} \sum_{i=1}^N \frac{\varphi(i) - \varrho(i)}{\max\{\varphi(i), \varrho(i)\}} \quad (5)$$

where $\varphi(i)$ is the average distance between vector i and vectors within the same cluster, while $\varrho(i)$ is the shortest average distance between vector i and vectors in other clusters. To simplify the interpretation of our framework, Table I lists commonly used notations along with their corresponding definitions.

IV. ALGORITHM DESIGN

In this section, we elaborate on the design of the WFL-VC algorithm.

A. WFL-VC Training Process

Intuitively, clustering clients requires the server to obtain individual information, which inherently contradicts the principles of AirFL. Moreover, when implementing TopK sparsification for compression, over-the-air aggregation becomes incompatible unless the indices of the top model updates' parameters are aligned prior to aggregation. To address these challenges, WFL-VC operates in two phases: Phase 1 focuses on client clustering, selection, and unifying local indices, while Phase 2 integrates the AirFL algorithm. Notably, the overhead of Phase 1 must be minimized. Otherwise, the benefits of OTA aggregation in Phase 2 would be negated. Therefore, in Phase 1, each model update is represented using only 1 bit.

To illustrate our algorithm, we present a simple example in Fig. 1. In Phase 1, each client transmits a 0–1 vote vector to the server, derived from locally executing the TopK compression. Each vote is determined by comparing the respective model update value with a threshold: it is set to 1 if the magnitude exceeds the threshold and 0 otherwise. Once these 0–1 votes are received, the PS clusters clients based on the distance between the collected vote vectors. Then, each cluster will be represented by a fixed number of clients selected based on both their distance from the cluster centroid and their selection probability. At the same time, the PS aggregates vote vectors to determine the indices of the global top model updates, which

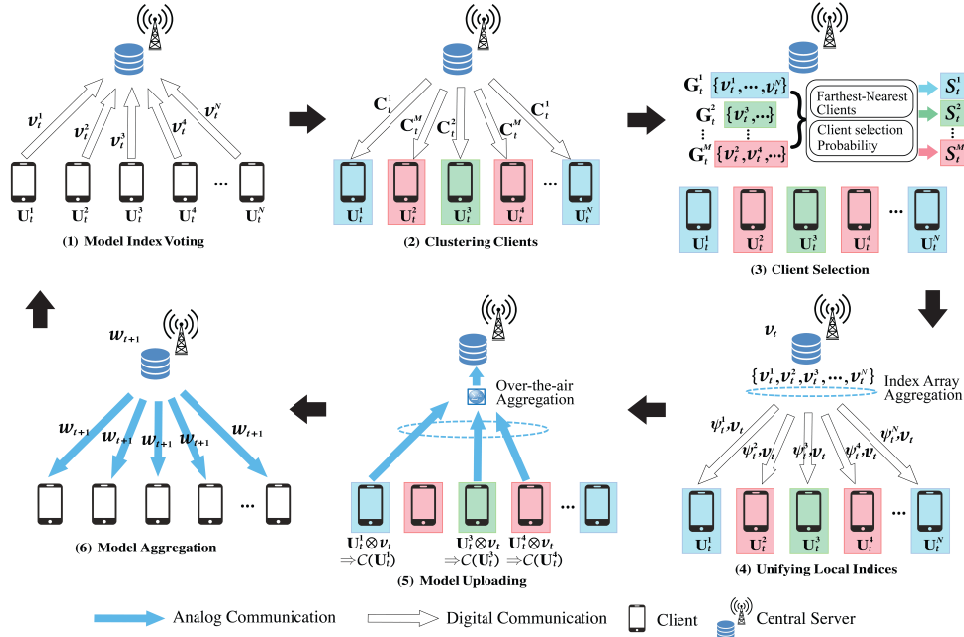


Fig. 1. Training process and cluster selection in WFL-VC.

are then distributed to all clients. In Phase 2, only selected clients will transmit their global top model updates to the PS via analog transmission. The detailed steps of WFL-VC in the t -th global iteration are as follows.

- 1) *Model Index Voting:* At the start of every iteration, client i synchronizes its local model $w_{t,0}^i$ with the latest global model w_t distributed by the PS. After E local training steps, the client computes its module update as the difference between the initial $w_{t,0}^i$ and trained model $w_{t,E}^i$. To compensate for prior model compression errors, this update is then adjusted by incorporating the residual error e_t^i from the previous global iteration. The final model update is thus: $U_t^i = (w_{t,0}^i - w_{t,E}^i) + e_t^i$. In order to cluster clients with minimal cost, all clients will vote for local significant model updates. Client i votes the top k local model updates of the largest magnitude as significant local model updates. The 0–1 voting vector v_t^i is then uploaded to the PS. Note that k can be proportional to the model dimension d .
- 2) *Clustering Clients:* The PS performs density-based clustering over the received vote vectors $\{v_t^1, \dots, v_t^N\}$ for grouping clients based on their similarity. Here, we temporarily omit the detailed clustering algorithm design, which will be explained further in Section IV-B1. We simply assume that clients will be grouped into M clusters, denoted by $C_t = \{C_t^1, C_t^2, \dots, C_t^m, \dots, C_t^M\}$. Meanwhile, let $\mathcal{G}_t = \{\mathcal{G}_t^1, \mathcal{G}_t^2, \dots, \mathcal{G}_t^m, \dots, \mathcal{G}_t^M\}$ represent the client IDs in each cluster. For example, the set $\mathcal{G}_t^1$ includes all client IDs in cluster 1.
- 3) *Clients Selection:* In WFL, transmitting model updates from clients to the PS can be energy-intensive. To reduce energy consumption without significantly compromising model accuracy, we propose a cluster-based client selection approach motivated by two key observations: 1) clients from different clusters can complement each

other to jointly improve the learning performance and 2) careful selection can maintain model accuracy with reduced energy costs. Our approach selects only a small fixed number of representative clients (e.g., n per cluster) for transmitting model updates. The selection criteria are also detailed in Section IV-B2.

- 4) *Unify Model Indices:* Since the set of selected clients $S_t = \{S_t^1, S_t^2, \dots, S_t^M\}$ varies at each iteration, estimating the global indices vector using only these clients results in fluctuations in the vector's length. Thus, the PS aggregates 0–1 index arrays of all participating clients to estimate the global top model updates $v_t = \sum_{i=1}^N v_t^i$. As v_t contains d elements, each element denoted by v_t^l for $l = 1, \dots, d$. After the index array aggregation, the PS applies a threshold integer a to eliminate insignificant model updates in v_t . Thus, if $v_t^l < a$, WFL-VC sets $v_t^l = 0$ or otherwise; if $v_t^l \geq a$, WFL-VC sets $v_t^l = 1$. A particular dimension is deemed significant only if it receives votes from at least a clients. The finalized v_t acts as the global index array, which the PS distributes back to clients.
- 5) *Model Uploading:* Upon receiving v_t , selected clients upload their model updates following the global index vector as $C(U_t^i)$. In this sparse model updates obtained based on the global index vector v_t , only model updates' parameters aligned with the nonzero elements in the global index vector are uploaded. For robust training, in each global iteration the model updates that are not selected as significant model updates are accumulated as the residual error $e_t^i = C(U_{t-1}^i) - U_{t-1}^i$ [44], [45], [46]. Previous studies have demonstrated that incorporating the residual error improves the convergence performance for cases with high model compression ratios [44]. This approach enables achieving the same convergence rate as the classical uncompressed distributed SGD with

linear speedup [45]. Due to the wireless nature of the transmission, AirFL is susceptible to errors caused by channel fading and noise when transmitting the model update to the PS. To combat the effect of channel fading, client i precodes its model updates with a power scaling factor $p_{i,2}$. Additionally, to minimize the effect of channel noise, clients send only the model update values to the PS, which then applies the learning rate η_t to the aggregated model updates. Thus, client i actually transmits $(\sqrt{p_{i,2}}\mathcal{C}(\mathbf{U}_t^i)/\eta_t)$.

- 6) *Model Aggregation*: When the selected clients in $\mathcal{S}_t$ concurrently transmit their model updates in an analog fashion, the transmitted waveform symbols superimpose directly, completing model update aggregation. Therefore, the received aggregated signal at PS can be written as

$$\mathbf{U}_t = \sum_{i \in \mathcal{S}_t} h_i \frac{\sqrt{p_{i,2}}\mathcal{C}(\mathbf{U}_t^i)}{\eta_t} + \mathbf{z}_t \quad (6)$$

where h_i is the channel fading coefficient between client i and the PS, $p_{i,2}$ is the power scaling factor of client i in Phase 2, and $\mathbf{z}_t \sim \mathcal{N}(0, \delta^2 \mathbf{I})$ is the AWGN vector. Based on (6), the PS refines the global model as follows:

$$\begin{aligned} \mathbf{w}_{t+1} &= \mathbf{w}_t - \frac{\eta_t}{|\mathcal{S}_t|\epsilon} \mathbf{U}_t \\ &= \mathbf{w}_t - \frac{1}{|\mathcal{S}_t|} \left(\sum_{i \in \mathcal{S}_t} \frac{h_i \sqrt{p_{i,2}}}{\epsilon} \mathcal{C}(\mathbf{U}_t^i) + \frac{\eta_t \mathbf{z}_t}{\epsilon} \right) \end{aligned} \quad (7)$$

where ϵ represents the denoising factor, which can be adjusted to reduce the impact of channel fading and AWGN.

To provide a comprehensive overview of the training process, the pseudocode is presented in Algorithm 1.

B. Client Clustering and Selection

1) *Clustering Clients*: To avoid excessive communication and power costs, we propose clustering based on the voting vectors transmitted by clients. The DBSCAN algorithm [47] is a density-based clustering method that groups clients based on the similarity measure (e.g., cosine similarity). It is used to cluster voting vectors to get the global data distribution. DBSCAN operates according to two key parameters: τ and ω , where τ represents the range of maximum distance between two clients to be considered neighbors and ω denotes the minimum number of clients in a τ -neighborhood required to form a dense region, or cluster. Since the number of clusters M is unknown, the DBSCAN parameters are selected using a heuristic approach to achieve optimal clustering. To determine the best clustering outcome, the average silhouette coefficient is computed for each clustering configuration, $\pi(\mathcal{C}_t)$. If any clients remain unclustered, the clusters' centroids are computed σ_t^m , and the remaining unclustered clients are assigned to the nearest cluster. Algorithm 2 outlines the pseudocode of the proposed clustering method.

Algorithm 1 Training Process of WFL-VC

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1: Input:  $\mathbf{w}_0$ ;  $N$ ;  $e_0^i = 0$ 
2: Output:  $\mathbf{w}_T$ 
3: for  $t = 0 \rightarrow T - 1$  do
4:   On client  $i$ :
5:   Initialize local model  $\mathbf{w}_{t,0}^i := \mathbf{w}_t$ .
6:   Update model  $\mathbf{w}_{t,j+1}^i$  as in Eq. (2) for  $E$  iterations to get  $\mathbf{w}_{t,E}^i$ .
7:   Compute model update  $\mathbf{U}_t^i = (\mathbf{w}_{t,0}^i - \mathbf{w}_{t,E}^i) + \mathbf{e}_t^i$ .
8:   // Model Index Voting
9:   Vote  $k$  elements from  $\mathbf{U}_t^i$  with probabilities proportional to their magnitude to obtain  $\mathbf{v}_t^i$ .
10:  Upload voting vector  $\mathbf{v}_t^i$  to PS via digital communication.
11:  Download the global index array  $\mathbf{v}_t$  and selected clients indicator  $\psi_t^i$  from PS.
12:  // Model Uploading
13:  Use  $\mathbf{v}_t$  to determine model updates to get  $\mathcal{C}(\mathbf{U}_t^i)$ .
14:  Update residual error  $\mathbf{e}_{t+1}^i = \mathcal{C}(\mathbf{U}_t^i) - \mathbf{U}_t^i$ .
15:  Transmit  $\mathcal{C}(\mathbf{U}_t^i)$  to the PS via analog communication after scaling with energy factor  $p_{i,2}$ .
16:  Download the updated global model  $\mathbf{w}_{t+1}$  from PS.
17:  On PS:
18:  // Clustering Clients
19:   $\mathcal{G}_t, \mathbf{C}_t \leftarrow \text{CLUSTERINGCLIENTS}(\mathbf{v}_t^1, \dots, \mathbf{v}_t^N)$ 
20:  // Client Selection
21:   $\mathcal{S}_t \leftarrow \text{CLIENTSELECTION}(\mathcal{G}_t, \mathbf{C}_t)$ 
22:  // Unifying Model Indices
23:  Aggregate clients' voting vectors to estimate the global index array  $\mathbf{v}_t$  based on the threshold  $a$ .
24:  Distribute global index array  $\mathbf{v}_t$  and client selection indicator  $\psi_t^i$  to clients.
25:  // Model Aggregation
26:  Aggregate transmitted signals from  $\mathcal{S}_t$  clients via over-the-air computation to obtain  $\mathbf{U}_t$  and update  $\mathbf{w}_t$  as in Eq. (7).
27:  Distribute updated  $\mathbf{w}_{t+1}$  to clients.
28: end for

```

2) *Clients Selection*: To enhance the diversity of training models contributing to global model updates, unselected clients from the last few global iterations should be prioritized in the current t -th global iteration, mitigating participant over-selection. Let ρ_t^i denote client i 's selection probability, initialized as $\rho_t^i = (1/N)$ for all i . This probability is updated dynamically as follows: if client i is selected in the current global iteration, then its selection probability is updated as

$$\rho_{t+1}^i = \lambda \rho_t^i, \quad \lambda \in (0, 1] \quad (8)$$

where λ is the selection probability decay factor. Meanwhile, for any unselected client j , the selection probability increases as follows:

$$\rho_{t+1}^j = \rho_t^j + (1 - \lambda) \rho_t^j. \quad (9)$$

In this work, we propose a client selection method that selects n clients (e.g., $|\mathcal{S}_t^m| = n$) to represent each cluster $\mathcal{C}_t^m$ based on two complementary criteria.

Algorithm 2 Clustering Clients

```

1: Input:  $\pi_{\max} = 0; \tau_{\min}; \tau_{\max}; \omega_{\min}; \omega_{\max}$ 
2: Output:  $\mathbf{C}_t; \mathcal{G}_t$ 
3: Calculate the cosine similarity  $\text{dis}(\mathbf{v}_t^i, \mathbf{v}_t^j)$  as in Eq. (4).
4: for  $\tau_i = \tau_{\min}, \dots, \tau_{\max}$  do
5:   for  $\omega_j = \omega_{\min}, \dots, \omega_{\max}$  do
6:      $(\mathbf{C}, \mathcal{G}) \leftarrow \text{DBSCAN}(\text{dis}(\mathbf{v}_t^i, \mathbf{v}_t^j), \tau_i, \omega_j)$ 
7:     Compute  $\pi(\mathbf{C})$  as in Eq. (5)
8:     if  $\pi(\mathbf{C}) > \pi_{\max}$  then
9:        $\pi_{\max} \leftarrow \pi(\mathbf{C})$ 
10:       $\mathbf{C}_t \leftarrow \mathbf{C}$ 
11:    end if
12:  end for
13: end for
14: Set clients group  $\mathcal{G}_t^m$  corresponding to cluster  $\mathbf{C}_t^m$ .
15: if some client vectors remain unclustered, i.e.,  $\mathbf{v}_t^x \in \{\mathbf{v}_t^1, \dots, \mathbf{v}_t^N\}$  then
16:   Calculate clusters centroids  $\mathbf{o}_t^m = \frac{1}{|\mathcal{G}_t^m|} \sum_{\mathbf{v} \in \mathcal{G}_t^m} \mathbf{v}$ 
17:   for each  $\mathbf{v}_t^j \in \mathbf{v}_t^x$  do
18:     for  $m = 1, 2, \dots, M$  do
19:       Calculate cosine similarity  $\text{dis}(\mathbf{v}_t^j, \mathbf{o}_t^m)$ .
20:     end for
21:     Assign  $\mathbf{v}_t^j$  to the cluster with the nearest centroid.
22:   end for
23:   Update client groups  $\mathcal{G}_t^m$  corresponding to clusters  $\mathbf{C}_t^m$ .
24: end if

```

Algorithm 3 Clients Selection

```

1: Input:  $\mathbf{C}_t; \mathcal{G}_t; b; c$ 
2: Output:  $\mathcal{S}_t$ 
3: for  $m = 1, 2, \dots, M$  do
4:   Formulate  $\rho^{\mathcal{G}_t^m}$ .
5:   Calculate distance-based priority criterion  $\chi_t^m$  as in Eq. (10).
6:   Calculate participant selection method  $\kappa_t^m$  as in Eq. (11).
7:    $\mathcal{S}_t^m \leftarrow \{\mathbf{v}_t^i, \mathbf{v}_t^j, \dots, \mathbf{v}_t^n \mid \kappa_t^m(i) \geq \kappa_t^m(j) > \dots > \kappa_t^m(n)\}$ 
8: end for
9: Update  $\{\rho_{t+1}^1, \dots, \rho_{t+1}^N\}$  using Eqs. (8), (9).

```

The first criterion is based on their distance from the cluster centroid $\mathbf{o}_t^m$: we favor those closest to or farthest from $\mathbf{o}_t^m$, while deprioritizing clients at intermediate distances. This approach aims to balance cluster representation by considering both central tendency and spread characteristics. The distance-based priority criterion can be represented using the U-shaped function as

$$\chi_t^m(i) = \frac{2d_i - (d_{\min} + d_{\max})}{d_{\max} - d_{\min}}, \quad i \in \mathcal{G}_t^m \quad (10)$$

where $d_{\max} = \arg \max_{i \in \mathcal{G}_t^m} \text{dis}(\mathbf{v}_t^i, \mathbf{o}_t^m)$, $d_i = \text{dis}(\mathbf{v}_t^i, \mathbf{o}_t^m)$, and $d_{\min} = \arg \min_{i \in \mathcal{G}_t^m} \text{dis}(\mathbf{v}_t^i, \mathbf{o}_t^m)$.

The second criterion is the client selection probability from the previous global iteration. Given the selection probability of all clients at the m -th group ($\mathcal{G}_t^m$) as $\rho^{\mathcal{G}_t^m}$, the participant selection method at the t -th global iteration for m -th cluster

($\mathbf{C}_t^m$) can be written as

$$\kappa_t^m = b * \rho^{\mathcal{G}_t^m} + c * \chi_t^m \quad (11)$$

where b and c are weighting parameters controlling each criterion's relative influence. Clients are first sorted by their scores in κ_t^m ; the top n clients with the highest scores are then selected to represent $\mathcal{G}_t^m$ as $\mathcal{S}_t^m = \{\mathbf{v}_t^i, \mathbf{v}_t^j, \dots, \mathbf{v}_t^n\}$, where $\kappa_t^m(i) \geq \kappa_t^m(j) > \dots > \kappa_t^m(n)$. The pseudocode of the proposed client selection method is demonstrated in Algorithm 3.

C. Transmission Model

This section investigates model accuracy optimization under a fixed training time through the calibration of the power factors $p_{i,1}, p_{i,2}$ and denoising factor ϵ .

Let Y denote the fixed training time span, the number of global iterations $T = (Y/y(\{p_{i,1}\}))$, where $y(\{p_{i,1}\})$ is the training time per global iteration. In WFL-VC, there are two stages for transmission. Thus, the training time can be modeled as follows:

$$y(\{p_{i,1}\}) = \tau_{\text{local}} + \tau_{\uparrow,1} + \tau_{\uparrow,2} + \tau_{\downarrow} \quad (12)$$

where τ_{local} denotes the time required to perform E numbers of local iterations, $\tau_{\uparrow,1}$ represents the uploading time of Phase 1, $\tau_{\uparrow,2}$ is the uploading time of Phase 2, and $\tau_{\downarrow}$ is the download time of global model updates and the deduced global votes. Next, we investigate the time duration of each transmission phase.

In Phase 1, clients transmit index arrays to the PS through digital communication, enabling the quantization and compression of the transmitted vectors. According to [48], the uploading rate of client i is expressed as: $\Omega_i = (W/N) \log_2(1 + (p_{i,1}h_i^2/\delta^2))$, where W represents the total bandwidth that is equally split and assigned to N clients, $p_{i,1}$ is the power scaling factor of client i in Phase 1, and δ is the noise power of AWGN. Let Q denote the bit size of the uploaded 0–1 vector after encoding, then the uploading time for 0–1 vectors in Phase 1: $\tau_{\uparrow,1} = \max_{i \in N}(Q/\Omega_i)$.

In Phase 2, clients transmit model updates as analog symbols. Assuming the number of symbols to be transmitted is q , and the time slot for each resource block to transmit X symbols is denoted as T_{slot} . The uploading time for model updates in Phase 2 is $\tau_{\uparrow,2} = \lceil (q/X) \rceil * T_{\text{slot}}$, where $\lceil \cdot \rceil$ is the integer ceiling function. Given the maximum power budget $P_{\max,t}^i$ for client i at the t -th iteration. Then, the actual power consumption, $P_{\text{actual},t}^i$ is as follows:

$$p_{i,1} * \frac{Q}{\Omega_i} + \psi_t^i * p_{i,2} * \left\| \frac{\mathcal{C}(\mathbf{U}_t^i)}{\eta_t} \right\|^2 \leq P_{\max,t}^i \quad (13)$$

where $\psi_t^i \in \{0, 1\}$ indicates if the client i is selected or not for global model training.

Considering that the communication time in Phase 1 is dictated by the slowest client, the optimal strategy is for all N participants to adopt the transmission speed of the slowest client. We define $p_{i,1} * h_i^2 = \theta, \forall i \in N$, where $\theta_{\max} = \min_{i \in N} P_{\max,t}^i * h_i^2$. We traverse θ in the range $0 < \theta \leq \theta_{\max}$ that for given θ the power scaling factor in Phase 1 for client i is computed as $p_{i,1} = (\theta/h_i^2)$. Given $p_{i,1}$ each selected client

optimizes its power scaling factor in Phase 2 $p_{i,2}, \forall i \in \mathcal{S}_t$ based on the remaining energy. The optimization of both the power scaling factor $p_{i,2}$ and denoising factor ϵ in WFL-VC follows the approach proposed by Su et al. [18], subject to the power constraint specified in (13).

D. Overhead Cost With WFL-VC Implementation

We outline three key considerations for implementing the WFL-VC algorithm in practical WFL systems.

1) *How to Tune a* : Tuning the threshold value a is essential, yet it only needs to be performed once. We present how to set a in a lightweight manner in practice. In the first global iteration $t = 1$, each client uploads k out of d model updates to the PS via digital communication. Assuming that each model update is represented using ζ bits and each index requires $\log_2 d$ bits, the overall communication traffic in a system with N clients scales as $\mathcal{O}(N * k(\log_2 d + \zeta))$. The PS tunes a as a hyperparameter by evaluating several candidate values and selecting the optimal one. In the second global iteration $t = 2$, the PS no longer needs to compute a , as the same value will be used for all remaining iterations of the training process.

2) *Phase 1 Communication Overhead*: Compared with existing frameworks, Phase 1 in WFL-VC offers a distinctive advantage, though it incurs communication overhead. In practice, however, this overhead remains lightweight, as each model update is represented by a single bit. The communication overhead in Phase 1 is primarily driven by the volume of index vector traffic Q that each participating client must upload. For N participating clients, each uploading k indices of $\log_2 d$ bits, the communication overhead scales as $\mathcal{O}(N * k \log_2 d)$.

3) *Phase 1 Complexity Overhead*: The clustering in Phase 1 employs the DBSCAN algorithm, which incurs a computational complexity of $\mathcal{O}(N^2)$, where N is the number of participating clients. To better understand and measure the impact, we analyze in Section V-B4 how this added server-side computation compares with the resulting client-side energy savings.

V. PERFORMANCE EVALUATION

This section presents a simulation of a WFL system employing over-the-air computation for model aggregation. We conduct extensive experiments to compare the performance of our proposed framework against the state-of-the-art benchmarks and report the results in this section.

A. Experimental Settings

1) Datasets and Models:

1) *CIFAR-10*: Is a dataset containing 60 000 color images ($3 \times 32 \times 32$ pixels), with 50 000 designated for training and 10 000 for testing. The whole dataset is divided based on ten labels. We train a three-layer CNN for classification, with each layer structured as Convolution-BatchNormalization-MaxPooling. The model contains a total of 2 471 370 parameters and ends with three fully connected layers. We define the learning rate of

the training process as $(0.1/(1 + (\sqrt{t}/45)))$, where t represents the number of global iterations.

2) *MNIST*: Is a dataset of 60 000 grayscale images (28×28 pixels) designated for training and 10 000 images for testing. The dataset comprises ten labels of handwritten digits 0–9. We train a multilayer perceptron to classify MNIST images that consist of a 784-neuron input layer, a 64-neuron hidden layer, and a 10-neuron softmax output layer. The model includes 50 890 parameters. Similarly, we set the learning rate as $(0.2/(1 + \sqrt{t}))$. In our experiment, we distribute the training dataset among clients in a non-IID manner, where the label distribution for different clients is generated based on the Dirichlet distribution. The concentration parameter β , which controls the similarity among clients, is set to 0.1 to reflect the condition of high heterogeneity [49].

2) *System Settings*: In the WFL system, during each global iteration, clients perform local iterations using the Stochastic Gradient Descent (SGD) algorithm to update their local models, with a central PS coordinating the clients' collaboration to train a global model. We configure the system size with $N = 20$ clients, where each client performs $E = 5$ local iterations. The number of clients selected to represent each cluster is set to $n = 4$ clients. After completing local training, clients transform model updates into waveforms and transmit them to the PS for global model aggregation. The analog communication leverages the waveform superposition property of the wireless signals to perform over-the-air computing during transmission to achieve model aggregation.

In our simulation, similar to [18] and [27], the wireless channels between clients and the PS follow IID Rayleigh fading, i.e., h_i 's are generated from a Gaussian distribution $\mathcal{N}(\mu, \delta^2)$, where $\mu = 0$ and $\delta^2 = 1$. To calculate the elapsed communication time for transmitting analog symbols, we use $\lceil (q/X) \rceil * T_{\text{slot}}$ by referring to [18] and [27]. In LTE systems [50], each resource block has a duration $T_{\text{slot}} = 1$ ms and consists of two time slots with a length of $X = 14$ symbols. To account for communication asymmetry, we set the downlink speed for each client as 5 times its uplink speed. For digital communication, we set the bandwidth W to 1 MHz.

In WFL-VC, each client votes k local model updates, while the total number of finally selected indices is primarily determined by the voting threshold a . Accordingly, we set k to $2\%d$ in our experiments, and for threshold a , we enumerate 1–3. Based on our experimental results, we set $a = 2$ for the MNIST scenario and $a = 1$ for the CIFAR-10 scenario. The decay factor λ for the selection probability is set to 0.9, while the weight parameters b and c are assigned values of 0.9 and 0.1, respectively. Based on our experimental performance configuration, the per-iteration computation time for image classification training is 0.2 s for CIFAR-10 and 0.03 s for MNIST.

In WFL-VTopK [18], all clients are assumed to have identical total power budget $P_{\text{total}} = 10d$, as illustrated in Fig. 2 (left). To highlight the impact of heterogeneous power constraints, we extend this setup by randomly assigning five distinct power budgets $P_{\text{total}} \in \{0.2, 0.4, 0.6, 0.8, 1\} \times 10d$ across

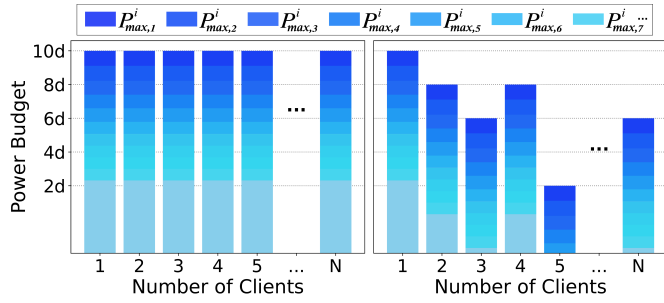


Fig. 2. Total power budget P_{total} for clients, unified total power budget (left), and different total power budget (right).

clients, as shown in Fig. 2 (right). For a fair comparison with other baselines, we assume all clients use the same maximum power $P_{\text{max},t}^i$ during the same iteration, despite their varying total power budgets. At each communication round, the total power budget is updated iteratively according to $P_{\text{total}}^i = P_{\text{total}}^i - P_{\text{actual},t-1}^i$. Clients are excluded from the system when their power is depleted.

3) *DBSCAN Configurations*: The DBSCAN clustering hyperparameters τ and ω primarily depend on the structure and density distribution of the clients' model updates. As a result, the optimal DBSCAN hyperparameter values vary across different datasets. When τ or ω is set improperly, clustering performance deteriorates. A small τ may cause clients that should belong to the same cluster to be incorrectly classified into different clusters, whereas a large τ can merge distinct clusters. Likewise, a small ω makes DBSCAN overly permissive, allowing small dense groups of clients to form clusters and potentially incorporating noise, while an excessively large ω imposes strict density requirements that suppress valid clusters and label many clients as noise. However, to the best of our knowledge, there is no theoretical guidance for selecting the parameters τ and ω , which means choosing appropriate parameters relies heavily on empirical tuning and extensive experimentation [51]. Several approaches have been proposed, where the choice of the appropriate approach depends on the dataset characteristics and clustering performance requirements [52]. To ensure adaptability across datasets and reduce the computation overhead of determining the initial exploration range for τ , we propose computing the inter-client similarity distances at the first global iteration and defining $\tau_{\min}$ and $\tau_{\max}$ using the minimum and maximum observed distances, respectively. While the minimum number of neighbors ω is set within the range $[2, (N/2)]$.

4) *Baselines*: We evaluate our work by implementing the following state-of-the-art WFL algorithms for comparison.

- 1) *FedAvg-OTA*: Represents the conventional WFL that applies FedAvg using over-the-air computation without optimizing the transmission power or denoising factor.
- 2) *Air-FedAvg* [27]: Proposes a power control scheme designed to address power constraints in WFL with over-the-air computation, aiming to minimize aggregation error. It further analyzes the impact of power and denoising factors on optimizing FL performance.

- 3) *WFL-VTopK* [18]: Is a WFL with model compression and power allocation schemes. It employs a voting-based algorithm for compression that aligns TopK sparsified model updates, facilitating direct over-the-air computation. At each iteration, a specified number of clients S are randomly selected to send (k/S) indices of their local significant model updates to the PS. These are then aggregated to form global significant model updates, which all clients will use to send their aligned model update parameters. It suggests allocating power for both digital and analog transmissions, which transmit indices and model updates, respectively.

- 4) *OBCSAA* [33]: Adopts compressive and relies on over-the-air computing for model aggregation. In OBCSAA, model updates undergo multiple compression stages: first, the TopK-sparsification strategy is applied; second, a measurement matrix $\Phi \in \mathbb{R}^{d \times d}$ is used to map model updates, reducing their dimensions from d -dimension to $\hat{d}$ -dimension; third, each model update parameter is quantized to 1-bit. In our experiments, the sparsity factor in OBCSAA is set to $k = 1\%d$, and the model updates reduction dimension is set to $\hat{d} = k*N$, which satisfies the restricted isometry property (RIP) condition, ensuring reliable reconstruction of the compressed gradients.

B. Experimental Results

We evaluate our proposed frameworks under fixed training time span $Y = 30$ and 5000 s on MNIST and CIFAR-10, respectively. We analyze the training convergence accuracy and the accumulation of power consumption over the training time span, as illustrated in Figs. 3 and 4. Furthermore, as the simulation considers both uncompressed and compressed frameworks, the convergence accuracy is further examined across communication rounds, as presented in Figs. 5 and 6. In this work, we assume clients have different power budgets, leaving some vulnerable to power depletion. Therefore, we monitor the number of clients with sufficient power to participate in the model training process over the designated time span.

1) *Uncompressed Transmission Frameworks*: Our experiments show that FedAvg-OTA achieves no improvement in training accuracy despite its high power consumption. This is due to the lack of effective power and denoising optimization, which degrades transmission quality. As training progresses, variation in client power budgets may cause some clients to run out of power, dropping the number of participants and degrading overall performance. Air-FedAvg outperforms FedAvg-OTA by using more effective power scaling and denoising, which better mitigates the model aggregation error introduced by channel noise. This leads to more accurate model updates and improved overall performance. However, this improvement comes at the cost of higher power consumption, as it requires full client participation throughout the training process. In the baselines discussed earlier that transmit uncompressed model updates, all model update parameters are sent to the PS. This exhausts the allocated time span, limiting the number of communication rounds to 6 for

TABLE II
PERFORMANCE EVALUATION OF WFL-VC FRAMEWORK

Framework	MNIST		CIFAR-10	
	Accuracy	Power Consumption	Accuracy	Power Consumption
FedAvg-OTA	8.92%	6.53×10^6	10.00%	2.91×10^8
Air-FedAvg	34.36%	5.01×10^6	25.32%	1.92×10^8
OBCSAA	30.55%	3.29×10^6	-	-
WFL-VTopK	62.64%	5.13×10^6	67.27%	1.20×10^8
WFL-VC	82.70%	1.22×10^6	67.15%	1.01×10^8

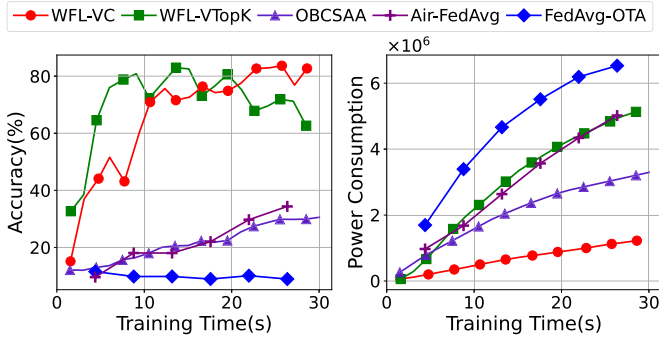


Fig. 3. Comparison between different algorithms in MNIST across training time, accuracy (left), and power consumption (right).

MNIST and 23 for CIFAR-10, as shown in Figs. 5 and 6, respectively.

2) *Compressed Transmission Frameworks*: Our experiments demonstrate that compression-based transmission frameworks achieve significantly higher accuracy over uncompressed transmission frameworks. Reduced transmission load shortens the global iteration time and enables the system to complete more communication rounds, thereby accelerating convergence.

By limiting the number of clients participating in model training and compressing model updates, WFL-VC significantly reduces power consumption without compromising model training accuracy, as demonstrated in Figs. 3 and 4. Table II reveals that WFL-VC achieves a 32.02% improvement in model accuracy while reducing power consumption by 76.21% in MNIST. For CIFAR-10, WFL-VC maintains baseline-equivalent accuracy while reducing power needs by 15.83%.

In OBCSAA, as evaluated on the MNIST dataset, the power usage per iteration remains independent of the magnitude of the client model gradients, leading to a decline in total power in later communication rounds as fewer clients participate. The compression approach in OBCSAA applies a d^2 dimension transformation matrix to compress the transmitted model updates, resulting in a computational complexity of $\mathcal{O}(d^2)$. Consequently, this approach introduces significant scalability challenges, as its computational load becomes increasingly burdensome with model size. For example, with CIFAR-10, where $d \approx 10^6$, this implies that over 4.5 TB of memory would be required for the transformation operation, which is extremely impractical. This high computational and memory overhead makes applying OBCSAA under limited

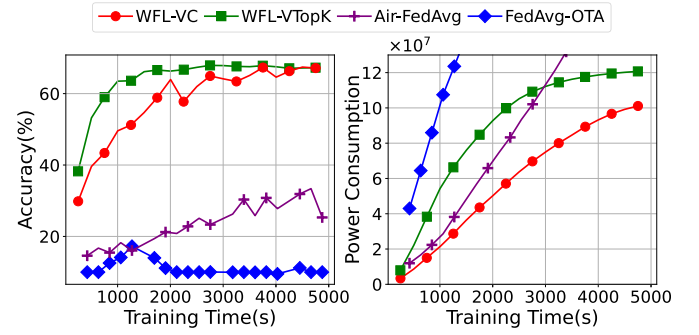


Fig. 4. Comparison between different algorithms in CIFAR-10 across training time, accuracy (left), and power consumption (right).

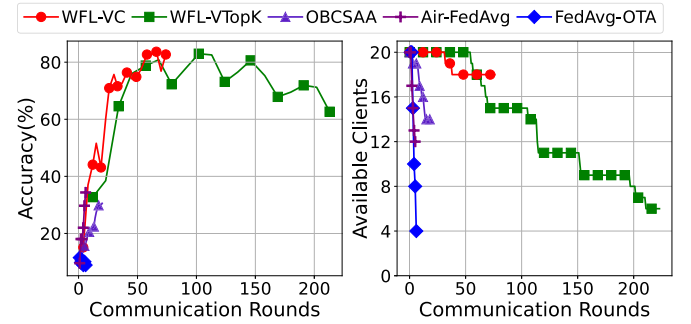


Fig. 5. Comparison between different algorithms in MNIST across communication rounds, accuracy (left), and number of available clients (right).

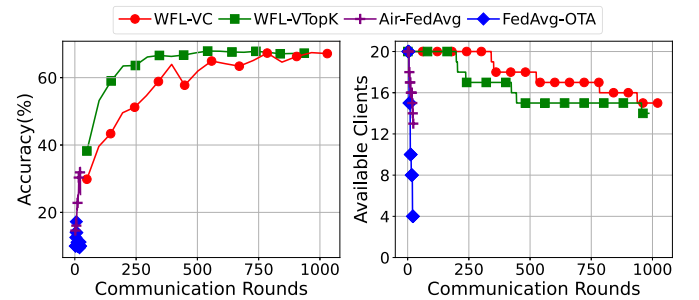


Fig. 6. Comparison between different algorithms in CIFAR-10 across communication rounds, accuracy (left), and number of available clients (right).

resource conditions unsuitable for training high-dimensional models (i.e., >2 million parameters), such as in classifying CIFAR-10 images [18].

Figs. 3 (right) and 4 (right) show that WFL-VTopK and WFL-VC exhibit a similar power consumption trend. As training commences, both begin with residual error $e_i^0 = 0$, which then accumulates unselected parameters during training, resulting in linear power consumption growth. As training converges toward peak performance, the difference between consecutive models decreases [53], shrinking the magnitude of the transmitted model gradients. Since model gradient transmission dominates power consumption, this drop in size reduces the system's power consumption. Despite exhibiting similar power consumption patterns, WFL-VC outperforms WFL-VTopK in power efficiency on both MNIST and

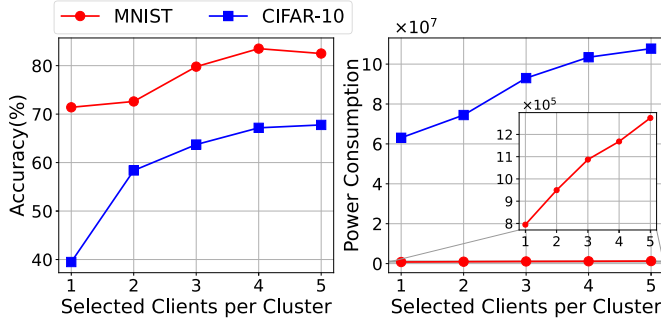


Fig. 7. Comparison between different numbers of selected clients to represent clusters in MNIST and CIFAR-10, accuracy (left), and power consumption (right).

CIFAR-10 datasets, attributable to its optimized client selection strategy.

WFL-VC's comprehensive voting mechanism fundamentally differs from WFL-VTopK's selective approach. In WFL-VTopK, the number of clients participating in voting was set to 5 randomly selected clients, with each voting $(k/5)$ elements. WFL-VC instead requires each active client to vote k elements, resulting in longer global top model update $\mathbf{v}_t$, larger local model transmissions, and increased upload/download latencies $(\tau_{\uparrow,1}, \tau_{\uparrow,2}, \tau_{\downarrow})$. Consequently, within a fixed training time span, WFL-VC completes fewer communication rounds than WFL-VTopK, as shown in Fig. 5 (left) for the MNIST dataset. Facing a $50\times$ larger model size in CIFAR-10 compared to MNIST, WFL-VC addresses potential uploading and downloading delays through scaling down voting elements from k to (k/N) . While WFL-VTopK's limited five-client voting pool minimizes power impact from poor channel conditions in Phase 1 transmission, WFL-VC's universal voting participation exposes it to substantial power drains from clients with poor channel conditions. We address this by implementing a convergence-aware termination criterion for the joint optimization problem $\mathbb{P}1 : \min \mathcal{J}(\epsilon, \{p_{i,1}\}, \{p_{i,2}\})$ in [18], preventing excessive power costs during $p_{i,1}$ refinement for marginal gains, e.g., $\Delta \mathcal{J} < 1$.

It is worth noting that the proposed client clustering method, implemented using the DBSCAN algorithm, does not control the resulting clusters' size. Notably, some clusters may contain as few as two or three clients. When a cluster m has $|\mathcal{G}_t^m| \leq n$ clients, we select all clients in the group $\mathcal{G}_t^m$ to represent $\mathcal{C}_t^m$.

3) *Robustness Evaluation*: The robustness of WFL-VC is evaluated by analyzing its performance under various configurations of key parameters. We examine the impact of different client selection ranges used to represent clusters during the training phase on the model accuracy, as illustrated in Fig. 7. The results indicate that increasing the number of selected clients to represent each cluster enhances convergence. However, this improvement comes at the expense of higher power consumption.

In Fig. 8, we tune the weighting parameters (b, c) values from $(0.1, 0.9)$ to $(0.9, 0.1)$ and measure the realized model accuracy. Since the weighting parameters b and c govern how clients are prioritized for cluster representation, they may affect the training model's convergence accuracy. However,

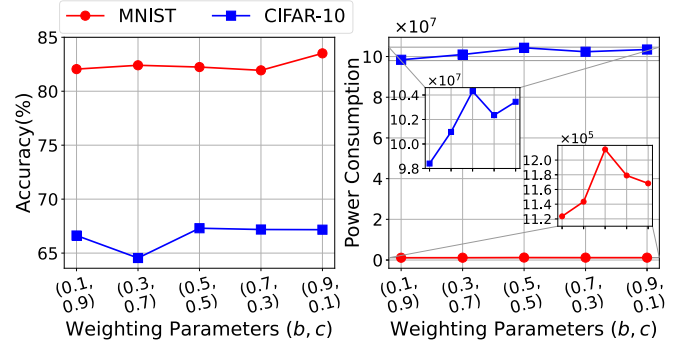


Fig. 8. Comparison between different set of hyperparameters for b and c values in MNIST and CIFAR-10, accuracy (left), and power consumption (right).

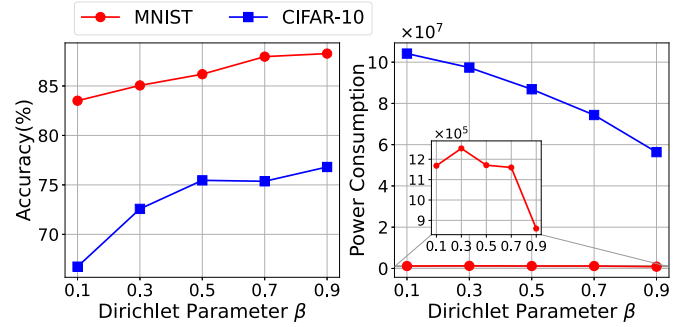


Fig. 9. Comparison between different non-IID data degrees in MNIST and CIFAR-10, accuracy (left), and power consumption (right).

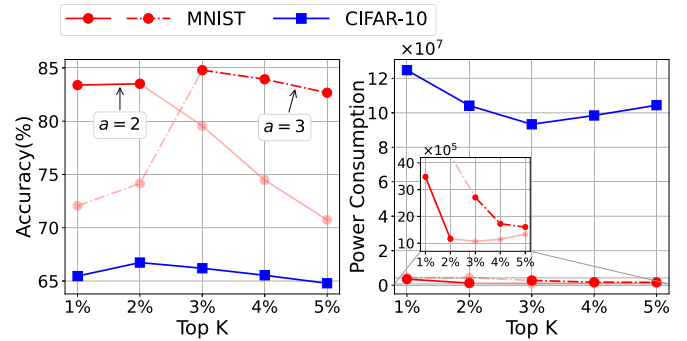


Fig. 10. Comparison between different k rates in MNIST and CIFAR-10, accuracy (left), and power consumption (right).

the empirical results show that the model's accuracy remains relatively stable, while power consumption variation is limited to 8% for MNIST and 6% for CIFAR-10.

We further evaluate the performance of WFL-VC under different degrees of data heterogeneity by varying the Dirichlet distribution parameter β from 0.1 to 0.9. As depicted in Fig. 9, experiments on both MNIST and CIFAR-10 datasets show that model accuracy improves gradually as the degree of non-IID data decreases.

In Fig. 10, we vary the number of votes k from $1\%d$ to $5\%d$, with d representing the model dimension, to examine the model accuracy and power consumption. As stated before,

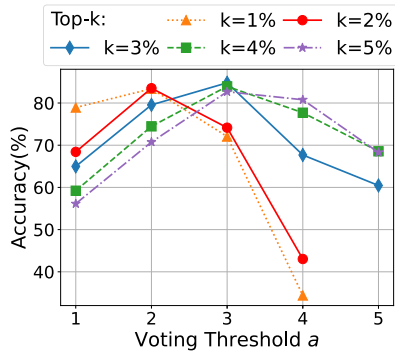


Fig. 11. Comparison between different voting threshold a values over a range of k rates in MNIST.

in the CIFAR-10 dataset experiment, the voting elements are scaled down to (k/N) . Hence, when the PS receives these votes from N clients for the index array aggregation, if a model parameter that receives at least one vote is treated as a global top model update, then the maximum possible length of the global index array $(k/N) * N$ is upper bounded by k of the model update parameters. Thus, simple OR aggregation is used to estimate global top model indices, similar to [18], in which $a = 1$ across all different k ranges as shown in Fig. 10. In contrast, for the MNIST dataset experiment, clients contribute k votes for the index array aggregation. Allowing the global index array to grow up to $k*N$ when OR aggregation is applied, which violates the intended sparsification rate k . To address this issue, the threshold a is treated as a hyperparameter to regulate the global index array length under specific k [54]. Using a fixed a across different sparsification rates leads to suboptimal behavior, it may become too large for small k , causing an excessive number of model updates to be filtered out. Conversely, for large k , the same threshold may be too lenient, resulting in an excessive number of model update indices, increasing traffic, and reducing communication rounds. This explains the framework performance on MNIST indicated by the faded lines in Fig. 10, where a higher threshold $a = 3$ is applied for small sparsification rates ($k = 1\%d - 2\%d$) and a lower threshold $a = 2$ is used for larger rates ($k = 3\%d - 5\%d$). Hence, we set $a = 2$ for $k = 1\%d - 2\%d$ and $a = 3$ for $k = 3\%d - 5\%d$.

We investigate the impact of varying sparsification rates k on WFL-VC performance for the MNIST dataset under threshold values $a = 1, 2, 3, 4, 5$. The experimental results exhibited in Fig. 11 show that each sparsification rate responds differently to changes in voting threshold a , which is consistent with the observations in Fig. 10.

4) *Overhead Cost*: As discussed in Section IV-D, the PS tunes a and performs client clustering and selection during Phase 1. Given that the server is a high-capacity computer equipped with an Intel Core i7-14700K CPU and an NVIDIA RTX 4090 GPU.

- 1) Tuning a in the first iteration incurs negligible overhead, while the associated communication overhead remains minimal, with upload latencies of 0.09 s for MNIST and 0.24 s for CIFAR-10. Likewise, the average power consumption per client remains extremely low, at

approximately 0.64% for MNIST and 0.04% for CIFAR-10, both of which are negligible in practice.

- 2) We report the computational cost of Phase 1 client clustering and selection as a percentage of total server computational time. Phase 1 accounts for 35.16% of the total computation in the MNIST experiment and 23.36% in the CIFAR-10 experiment.

VI. CONCLUSION

Despite the massive data collected by wireless devices such as IoT devices, harnessing this data is hindered by privacy concerns and limited resources associated with these devices. Recent studies have proposed AirFL with over-the-air aggregation as a promising method to enhance the communication efficiency of WFL. However, OTA aggregation conceals individual client information, rendering advanced FL algorithms that rely on such information ineffective in WFL. To address this challenge, we propose the WFL-VC algorithm, which integrates advanced client selection strategies and model compression with OTA aggregation. Our approach significantly reduces wireless device energy consumption while maintaining model accuracy. Extensive experiments on real-world datasets validate the effectiveness of our algorithms. Our findings underscore the potential of energy-aware WFL strategies in improving sustainability for resource-constrained wireless devices. Future work will focus on extending our approach to more heterogeneous and large-scale WFL environments.

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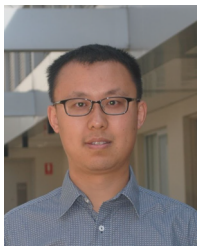
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